

Ziyad Sheebaelhamd

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EDUCATION

International Max Planck School for Intelligent Systems (University of Tübingen)

Tübingen, Germany

PhD student in Reinforcement Learning

September 2024 - Now

- Working on bridging the gap between Reinforcement Learning theory and practice under the supervision of Prof. Claire Vernade.
- Published my first project titled “Quantization-Free Autoregressive Action Transformer” at **NeurIPS 2025** as a [spotlight paper](#). Slides can be found [here](#).
- Published my second paper titled “[Split the Differences, Pool the Rest: Provably Efficient Multi-Objective Imitation](#)” at **NeurIPS 2026**.
- Submitted my third paper titled “Gaussian Distillation of Flow Policies” at **ICLR 2027**. Soon on arxiv.

Eidgenössische Technische Hochschule Zürich (ETH Zürich)

Zürich, Switzerland

Masters in Robotics Systems and Control | GPA: 5.84/6

December 2022

- **Master's Thesis:** Information Directed Reward Learning from Human Preferences in Reinforcement Learning. Supervised by Professor Andreas Krause. **Grade: 6/6**.
- **Semester Project:** A Nonlinear, Soft Constrained Model Predictive Safety Filter for Learning Based Controllers with Robust Stability Guarantees. Supervised by Professor Melanie Zeilinger. **Grade: 5.75/6**.
- **Published Paper at ICML Workshop, RL4RealLife:** [Safe Deep Reinforcement Learning for Multi-Agent Systems with Continuous Action Spaces](#)
- **Relevant Coursework:** Probabilistic Artificial Intelligence, Dynamic Programming and Optimal Control, Advanced Machine Learning, Deep Learning, Model Predictive Control, Recursive Estimation.

Karlsruher Institut für Technologie (KIT)

Karlsruhe, Germany

Bachelors in Mechanical Engineering | GPA: 1.7/1

June 2019

- **Bachelor's Thesis:** Bachelor Thesis Topic: Implementation of Object detection and localization for robotic grasping in an Autonomous Refrigeration System. **Grade: 1.3/1**
- **Awards:** Best grade certificate in Control Systems and Fluid Dynamics.

WORK EXPERIENCE

Merantix Momentum

Berlin, Germany

Machine Learning Engineer

09.2022 - 09.2024

- Developed and maintained an internal package to enable highly parallel image processing and model training using Flyte, Spark and Ray.
- Developed and deployed a pipeline to preprocess 5 Terabytes of satellite imagery to detect thin noise insulation walls in collaboration with the The German Center for Rail Traffic Research at the Federal Railway Authority
- Developed and deployed a demand forecasting solution and a fleet balancing algorithm for a taxi service in Germany using a Minimum Cost Flow problem formulation.
- Implemented a spark pipeline for a battery systems provider in order to enable free capacity trading.

Visium SA

Zürich, Switzerland

Junior Machine Learning (Internship)

09.2021 - 06.2022

- Causal discovery for an energy plant optimization using time series measurements.
- Large object detection in a 2500-square-meter waste field. Includes camera installation, data acquisition, data labeling and solution development.

Daimler AG**Sindelfingen, Germany***Production Planning Engineer (Internship)***02.2019 - 08.2019**

- Software development to establish message passing from robots' sensors to a data-visualization platform (Kibana) (MQTT, c#).
- Perform tests and data analytics on robot sensor outputs to assess cost reduction measures.

Karlsruher Institut für Technologie (KIT)**Karlsruhe, Germany***Teacher Assistant, Advanced Mathematics I/II***10.2016 - 10.2017**

- Holding weekly tutorials for students.

SKILLS & INTERESTS

Interests: Reinforcement Learning, Imitation Learning, Generative Modeling.**Programming Languages:** Python (Expert), C++ (Basic)**Stack:** Docker, Spark, Ray, Kubernetes, Flyte, Pytorch, Hydra.